POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5022 | Semester V | Theory | 4 Credits

Design of Machine Elements

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5022 as Design of Machine Elements in Semester V for Mechanical Engineering. The official SITTTR detailed source was unavailable. With user approval, this handbook uses MIT OpenCourseWare's Elements of Mechanical Design syllabus as a reliable supplementary source and does not represent recovered official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. This handbook cannot be used to fabricate or operate a component without current standards, verified loads/material data, tolerance/drawing review and qualified engineering approval.

Verified source note: The source supports bearings, springs, gears, mechanisms, modeling, design integration, fabrication/testing and responsible design practice.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Design of Machine Elements applies mechanics, materials, standards and design judgment to components such as joints, shafts, keys, couplings, bearings, springs and gears.

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□□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□
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Prerequisite foundation

Strength of materials, engineering mechanics, machine drawing and current design-code/data access.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain design process, loads, stress and failure considerations.
2. Apply conceptual design sequences to joints, shafts and couplings.
3. Explain selection/design principles for bearings, springs and gears.
4. Communicate assumptions, checks, manufacturability and safety limits of a design.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes		
CO	**Official outcome**	**Bloom level**
CO1	Explain machine-design process, loading, material and failure considerations.	Understand
CO2	Apply approved design-sequence concepts for joints, shafts, keys and couplings.	Apply
CO3	Explain gear, bearing and spring selection/design principles.	Understand
CO4	Document design assumptions, checks and limitations responsibly.	Apply

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7-PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Design Process, Loads and Materials	Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.	Approx. 14 hours	CO1	Module 1
2	Joints, Shafts and Couplings	Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.	Approx. 16 hours	CO2	Module 2
3	Gears, Bearings and Springs	Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.	Approx. 16 hours	CO3	Module 3
4	Integration, Testing and Responsibility	Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

CO1 · Approx. 14 hours

Joints, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

CO2 · Approx. 16 hours

Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

CO3 · Approx. 16 hours

Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

CO4 · Approx. 14 hours

MODULE 1

Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

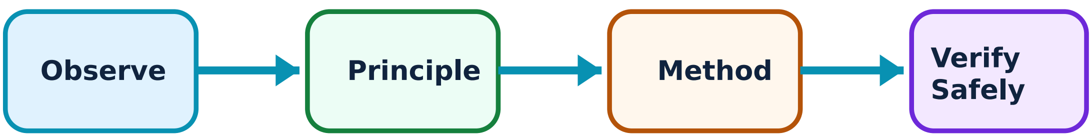
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Design Process, Loads and Materials: identify the system, state its principle, follow the correct method and verify the result safely.

1. Engineering design process−

A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review.

Study and application method

State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: A spreadsheet or CAD model is not validation without correct inputs and independent review.

2. Stress and failure awareness−

Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.

Study and application method

Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not combine incompatible material properties, units or safety factors.

3. Material and manufacturing choice−

Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection. Tolerances and surface finish affect fit and fatigue.

Study and application method

Compare candidates against stated criteria and record source/version of material data.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare candidates against stated criteria and record source/version of material data.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never substitute material or heat treatment without approved engineering change control.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Joints, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

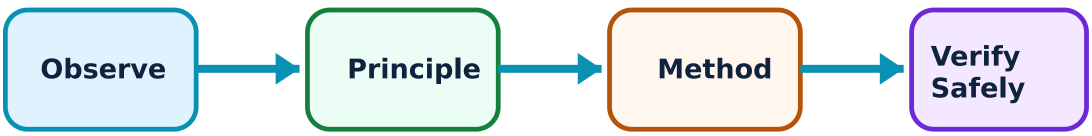
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Joints, Shafts and Couplings: identify the system, state its principle, follow the correct method and verify the result safely.

1. Joints and fasteners−

Bolted, welded and keyed joints transmit forces while allowing assembly/service. Preload, load distribution, fatigue, corrosion and inspection influence reliability.

Study and application method

Identify joint load path, failure modes and required design/inspection checks under the approved standard.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify joint load path, failure modes and required design/inspection checks under the approved standard.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use generic bolt torque or weld size without drawing, grade and procedure data.

2. Shafts, keys and couplings−

Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.

Study and application method

Determine torque/bending/load cases, critical geometry and prescribed equivalent-stress/twist checks; document keyway stress-concentration consideration.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Rotating machinery requires guarding, balance and qualified assembly; design exercise values are not operating approval.

3. Design verification−

Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.

Study and application method

Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not release a component before peer/qualified review.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

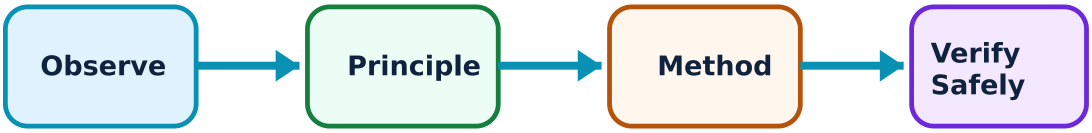
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Gears, Bearings and Springs: identify the system, state its principle, follow the correct method and verify the result safely.

1. Gears and drives−

Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.

Study and application method

State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not place hands near rotating drives; use guards and lockout before inspection.

2. Bearings and lubrication−

Bearings support radial/axial load while reducing friction. Selection depends on load, speed, life, alignment, lubrication, contamination and mounting.

Study and application method

Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wrong lubrication, preload or fit can rapidly damage a bearing.

3. Springs−

Springs store energy and provide force/deflection. Stress, fatigue, buckling, solid height and end condition are key considerations.

Study and application method

State load range, deflection, material, end condition and required checks under prescribed method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State load range, deflection, material, end condition and required checks under prescribed method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Compressed springs store energy; use controlled handling and guarding.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

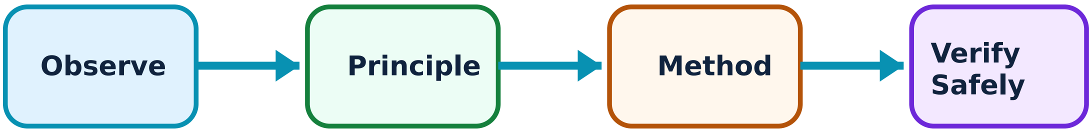
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Integration, Testing and Responsibility: identify the system, state its principle, follow the correct method and verify the result safely.

1. System integration−

Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.

Study and application method

Map interfaces and tolerance/assembly risks; plan inspection and functional test.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map interfaces and tolerance/assembly risks; plan inspection and functional test.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Local component adequacy does not guarantee system safety.

2. Testing and failure learning−

Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.

Study and application method

Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Never test beyond safe enclosure/load limits.

3. Design documentation−

Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.

Study and application method

Issue a complete controlled package with assumptions, units, revision and approval status.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Issue a complete controlled package with assumptions, units, revision and approval status.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not fabricate from uncontrolled or ambiguous drawings.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Design Process, Loads and Materials

Use the concept flow to revise observation, principle, method and verification.

Module 2: Joints, Shafts and Couplings

Use the concept flow to revise observation, principle, method and verification.

Module 3: Gears, Bearings and Springs

Use the concept flow to revise observation, principle, method and verification.

Module 4: Integration, Testing and Responsibility

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a load-path and failure-mode table for a shaft-coupling assembly.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare two bearing options using stated catalogue criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a design-review checklist for a spring or gear exercise.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Document a conceptual prototype test plan with safety controls.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Design Process, Loads and Materials

Explain Engineering design process and state one correct method, application or precaution.−

A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review.

Answer framework: State function, load cases, material/environment, life, safety factor basis, constraints and validation plan before calculation.

Important point: A spreadsheet or CAD model is not validation without correct inputs and independent review.

Explain Stress and failure awareness and state one correct method, application or precaution.−

Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.

Answer framework: Draw free-body/load path, identify critical section and relevant failure mode; use the prescribed theory/data with units.

Important point: Do not combine incompatible material properties, units or safety factors.

Explain Material and manufacturing choice and state one correct method, application or precaution.−

Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection. Tolerances and surface finish affect fit and fatigue.

Answer framework: Compare candidates against stated criteria and record source/version of material data.

Important point: Never substitute material or heat treatment without approved engineering change control.

Module 2 — Joints, Shafts and Couplings

Explain Joints and fasteners and state one correct method, application or precaution. −

Bolted, welded and keyed joints transmit forces while allowing assembly/service. Preload, load distribution, fatigue, corrosion and inspection influence reliability.

Answer framework: Identify joint load path, failure modes and required design/inspection checks under the approved standard.

Important point: Do not use generic bolt torque or weld size without drawing, grade and procedure data.

Explain Shafts, keys and couplings and state one correct method, application or precaution.−

Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.

Answer framework: Determine torque/bending/load cases, critical geometry and prescribed equivalent-stress/twist checks; document keyway stress-concentration consideration.

Important point: Rotating machinery requires guarding, balance and qualified assembly; design exercise values are not operating approval.

Explain Design verification and state one correct method, application or precaution.−

Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.

Answer framework: Create a checklist of assumptions, governing case, units, code/data source, result and independent sanity check.

Important point: Do not release a component before peer/qualified review.

Module 3 — Gears, Bearings and Springs

Explain Gears and drives and state one correct method, application or precaution.−

Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.

Answer framework: State ratio, power/speed, duty, load factor, material and required standard method; sketch force directions.

Important point: Do not place hands near rotating drives; use guards and lockout before inspection.

Explain Bearings and lubrication and state one correct method, application or precaution.−

Bearings support radial/axial load while reducing friction. Selection depends on load, speed, life, alignment, lubrication, contamination and mounting.

Answer framework: Identify bearing load/direction, operating environment and life requirement; follow catalogue/standard selection procedure.

Important point: Wrong lubrication, preload or fit can rapidly damage a bearing.

Explain Springs and state one correct method, application or precaution.−

Springs store energy and provide force/deflection. Stress, fatigue, buckling, solid height and end condition are key considerations.

Answer framework: State load range, deflection, material, end condition and required checks under prescribed method.

Important point: Compressed springs store energy; use controlled handling and guarding.

Module 4 — Integration, Testing and Responsibility

Explain System integration and state one correct method, application or precaution.−

Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.

Answer framework: Map interfaces and tolerance/assembly risks; plan inspection and functional test.

Important point: Local component adequacy does not guarantee system safety.

Explain Testing and failure learning and state one correct method, application or precaution.−

Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.

Answer framework: Define objective, instrumentation, acceptance criterion, controls and stop conditions before testing.

Important point: Never test beyond safe enclosure/load limits.

Explain Design documentation and state one correct method, application or precaution.−

Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.

Answer framework: Issue a complete controlled package with assumptions, units, revision and approval status.

Important point: Do not fabricate from uncontrolled or ambiguous drawings.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Design Process, Loads and Materials.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Joints, Shafts and Couplings.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Gears, Bearings and Springs.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Integration, Testing and Responsibility.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability..
2. Develop a complete answer or practical plan for: Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence..
3. Develop a complete answer or practical plan for: Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection..
4. Develop a complete answer or practical plan for: Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety..

LAST-MILE REVISION

Quick Revision

Module 1: Design Process, Loads and Materials

Design requirements, load paths, stress, fatigue, material properties, factor of safety and manufacturability.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Joints, Shafts and Couplings

Fasteners/welds, shafts/axles, keys, splines and couplings; loading, stress concentration and safe design sequence.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Gears, Bearings and Springs

Power transmission, gear geometry, rolling/plain bearings, lubrication, spring behaviour and selection.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Integration, Testing and Responsibility

Assemblies, tolerance stack-up, prototype/test, failure investigation, documentation and safety.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Engineering design process	A design begins with function, constraints, loads, environment and interfaces, then proceeds through modelling, selection, analysis, prototyping/testing and review.
Stress and failure awareness	Static, fluctuating, impact and thermal loads create normal/shear stress and potential yielding, fracture, fatigue, wear or buckling.
Material and manufacturing choice	Selection balances strength, stiffness, toughness, corrosion, cost, availability, process and inspection.
Joints and fasteners	Bolted, welded and keyed joints transmit forces while allowing assembly/service.
Shafts, keys and couplings	Shafts transmit torque often with bending; keys/couplings transfer torque and must avoid fretting, fatigue and misalignment problems.
Design verification	Design verification cross-checks load, stress, deflection, critical speed, fits, assembly and test plan against requirements.
Gears and drives	Gears transmit motion/torque with defined ratio; tooth strength, wear, alignment, lubrication and housing stiffness affect life/noise.
Bearings and lubrication	Bearings support radial/axial load while reducing friction.
Springs	Springs store energy and provide force/deflection.
System integration	Machine elements interact through interfaces, fits, stiffness, thermal expansion, lubrication and assembly sequence.
Testing and failure learning	Tests validate assumptions under controlled conditions; failure analysis identifies mechanism and corrective action.
Design documentation	Drawings, calculations, BOM, specifications, revisions and test records preserve design intent and traceability.

LEARNING RESOURCES

References

Recommended machine-element-design references

1. Joseph Shigley et al., Mechanical Engineering Design.
2. R. L. Norton, Design of Machinery.
3. Machinery's Handbook.

Approved public machine-design source

- https://ocw.mit.edu/courses/2-72-elements-of-mechanical-design-spring-2009/pages/syllabus/

This source supports the study handbook while official Course 5022 detail is unavailable; it does not replace standards or qualified design review.